

--	--	--	--	--	--	--	--	--

Student Number

2016

HALF YEARLY EXAMINATION

Extension 1 Mathematics

General Instructions

- Reading time – 5 minutes
- Working time - 2 hours
- Write using blue or black pen
Black pen is preferred
- Approved calculators may be used
- A table of standard integrals is provided at the back of this paper
- In Questions 11-14 show relevant mathematical reasoning and/or calculations
- Start a new booklet for each question

Total Marks – 70

Section I - Pages ****

10 marks

- Attempt Questions 1 – 10
- Allow about 15 minutes for this section

Section II - Pages ****

60 marks

- Attempt Questions 11 – 14
- Allow about 1 hour and 45 minutes for this section

Question	Marks
1 - 10	/10
11	/15
12	/15
13	/15
14	/15
Total	/70

THIS QUESTION PAPER MUST NOT BE REMOVED FROM THE EXAMINATION ROOM

This assessment task constitutes 30% of the Higher School Certificate Course Assessment

BLANK PAGE

Section I

10 marks

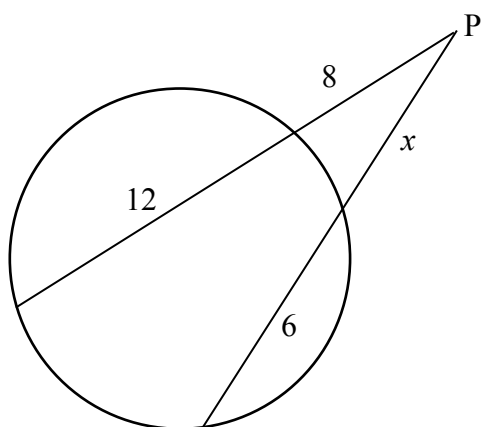
Attempt Questions 1 – 10

Allow about 15 minutes for this section

Use the multiple-choice answer sheet for questions 1 – 10 (Detach from paper)

1

Two secants from point P intersect a circle as shown in the diagram.



NOT TO SCALE

What is the value of x ?

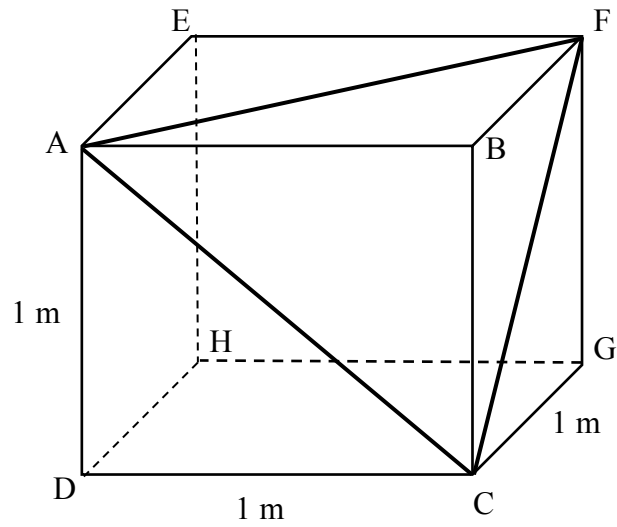
- (A) 4
- (B) 9
- (C) 10
- (D) 14

2

Which of the following is an expression for $\cos^6 x - \sin^6 x$?

- (A) $\sin 2x (\cos^4 x - \cos^2 x \sin^2 x + \sin^4 x)$
- (B) $\sin 2x (\cos^4 x + \cos^2 x \sin^2 x + \sin^4 x)$
- (C) $\cos 2x (\cos^4 x - \cos^2 x \sin^2 x + \sin^4 x)$
- (D) $\cos 2x (\cos^4 x + \cos^2 x \sin^2 x + \sin^4 x)$

3



The length of each edge of the cube ABCDEFGH is 1 metre.

What is the exact area of $\triangle ACF$?

- (A) $\frac{1}{2} \text{ m}^2$
- (B) $\frac{1}{\sqrt{2}} \text{ m}^2$
- (C) $\frac{\sqrt{3}}{2} \text{ m}^2$
- (D) 1 m^2

4 Which of the following is a solution to the equation $2^{2x} = 5$?

(A) $\frac{5}{4}$

(B) $\frac{\ln 5}{2 \ln 2}$

(C) $\frac{2 \ln 5}{\ln 2}$

(D) $\frac{\ln 5}{2}$

5 After using the substitution $u = 9 - x^2$, the integral

$$\int_0^1 6x\sqrt{9-x^2} \, dx$$

is rewritten as:

(A) $\int_0^1 -3\sqrt{u} \, du$

(B) $\int_0^1 -\frac{1}{3}\sqrt{u} \, du$

(C) $\int_9^8 -3\sqrt{u} \, du$

(D) $\int_9^8 -\frac{1}{3}\sqrt{u} \, du$

- 6 What is the correct solution to the inequality $\frac{x+1}{x+3} \leq -2$?
- (A) $-3 \leq x \leq -\frac{7}{3}$
- (B) $-3 < x \leq -\frac{7}{3}$
- (C) $x \leq -\frac{7}{3}$
- (D) $x \leq -\frac{7}{3}, x \neq -3$
- 7 What is the size of the acute angle between the lines $y = 2x - 5$ and $y = 4 - x$?
- (A) $71^{\circ}34'$
- (B) $18^{\circ}26'$
- (C) $40^{\circ}36'$
- (D) $33^{\circ}41'$
- 8 The interval between the 2 points A (-1, -3) and B (4, 12) is divided internally in the ratio 3:2 by the point C. The coordinates of point C are:
- (A) (1, 3)
- (B) $(-2\frac{2}{5}, -8\frac{2}{5})$
- (C) (2, 6)
- (D) $(2\frac{2}{5}, 8\frac{2}{5})$
- 9 The solutions to the equation $\cos 2\theta = \sin \theta$ where $0 \leq \theta \leq \pi$ are :
- (A) $\theta = \frac{\pi}{4}, \frac{3\pi}{4}$
- (B) $\theta = \frac{\pi}{12}, \frac{11\pi}{12}$
- (C) $\theta = \frac{\pi}{3}, \frac{2\pi}{3}$
- (D) $\theta = \frac{\pi}{6}, \frac{5\pi}{6}$

- 10** The correct expression of $\sin 2t + \sqrt{3}\cos 2t$ in the form $R\sin(2t + \alpha)$, where α is in radians is:
- (A) $2 \sin\left(2t + \frac{\pi}{3}\right)$
- (B) $\sqrt{2} \sin\left(2t + \frac{\pi}{3}\right)$
- (C) $2 \sin\left(2t + \frac{\pi}{6}\right)$
- (D) $\sqrt{2} \sin\left(2t + \frac{\pi}{6}\right)$

Section II

70 marks

Attempt Questions 11 – 14

Allow about 1 hours and 45 minutes for this section

Answer each question in a SEPARATE writing booklet. Extra writing booklets are available.

In Questions 11 – 14, your responses should include relevant mathematical reasoning and/or calculations.

Question 11(15 marks) Use a SEPARATE writing booklet

(a) Simplify $e^{\ln y}$ 1

(b) Find the range of the function $f(x) = \ln(x^2 + e^2)$. 2

(c) Differentiate $\frac{e^x}{x}$ 2

(d) Differentiate $\log_5 x^3$ with respect to x . 2

(e) A series is given as

$$S_n = \sin^2 x + 2\sin^4 x + 4\sin^6 x + \dots \quad \text{for } 0 < x < \pi$$

(i) Find the exact values of x for which the series has a limiting sum. 3

(ii) Show that the limiting sum of the series can be written in the form 2

$$\frac{\sin^2 x}{\cos 2x}$$

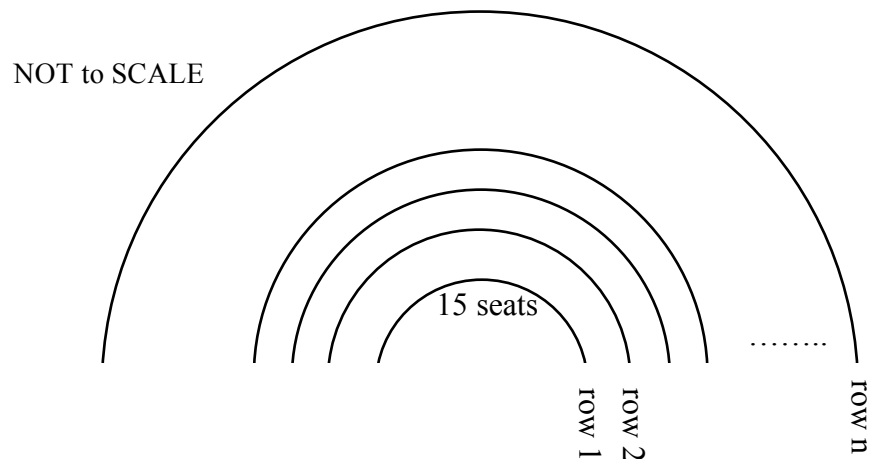
Question 11 continues on page 9

Question 11 (continued)

- (f) Seats in a semi-circular shaped theatre are arranged in rows.

The first row has 15 seats in it. Each row after the first row has 4 more seats than the row directly in front of it.

There are a total of n rows of seats in the theatre.



- (i) Find an expression for the total number of seats in the theatre. 1
- (ii) Harriet notices that there is an average of 43 seats per row.
How many seats are there in the last row of the theatre? 2

End of Question 11

Question 12 (15 marks) Use a SEPARATE writing booklet

(a) Using the substitution $u = \ln x$, or otherwise, find $\int \frac{1}{x(\ln x)^4} dx$, 3

(b) By using the substitution $x = \sin t$

(i) Show that $\int_0^{\frac{1}{2}} \sqrt{1-x^2} dx$ becomes $\int_0^{\frac{\pi}{6}} \cos^2 t dt$ 2

(ii) Hence evaluate $\int_0^{\frac{1}{2}} \sqrt{1-x^2} dx$ 3

(c)

(i) Starting from $\cos 3\theta = \cos(\theta + 2\theta)$ show that $\cos 3\theta = 4\cos^3 \theta - 3\cos \theta$ 2

(ii) Hence find $\int 12\cos^3 \theta - 9\cos \theta d\theta$ 2

(d) Using the substitution $u = 1 - 3x$, find $\int x\sqrt{1-3x} dx$. You can assume $x \leq \frac{1}{3}$. 3

End of Question 12

Question 13 (15 marks) Use a SEPARATE writing booklet

(a) Consider the function $f(x) = \frac{2x^2}{9-x^2}$

(i) Write down the equations of the asymptotes of $f(x)$. 2

(ii) Show that $f(x)$ is even. 1

(iii) Find any stationary points and determine their nature. 2

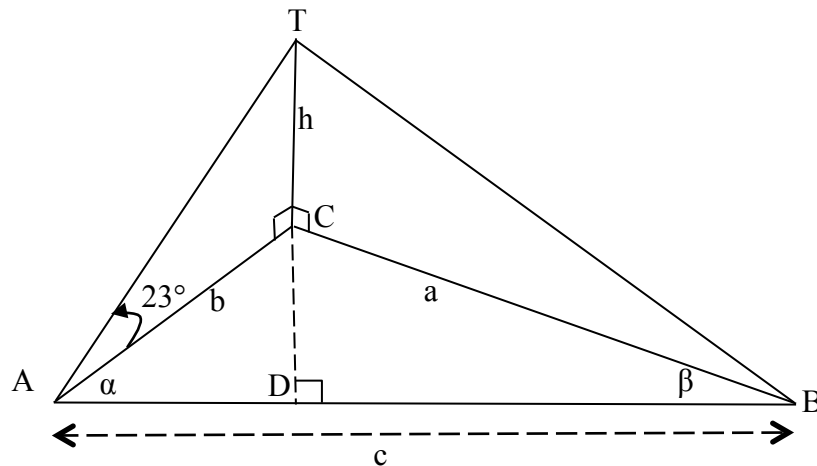
(iv) Sketch a graph of $f(x)$, showing all important features. The graph must be at least one third of a page in size. 2

Question 13 continues on page 12

Question 13 (continued)

(b)

NOT to SCALE



Triangle ΔABC lies on flat ground. At point C, a tower rises to a height of h metres. Angle $\angle CDB = 90^\circ$.

Let $\angle CAD = \alpha$ and $\angle CBD = \beta$.

- (i) Show that $a \sin \beta = b \sin \alpha$ 1
- (ii) Show that $c = a \cos \beta + b \cos \alpha$ 1
- (iii) Given that $c^2 = 4ab \cos \alpha \cos \beta$, show that $a = b$ 2
- (iv) Given that Point D is due South of the base of the tower and that the bearing of the tower from point B is 320° , explain why $\angle ACB = 80^\circ$. 1
- (v) Given that at point A, the angle of elevation of the top of the tower, T, is 23° and that distance $AB = 85$ metres, calculate the height, h , of the tower. Answer correct to 1 decimal place. 3

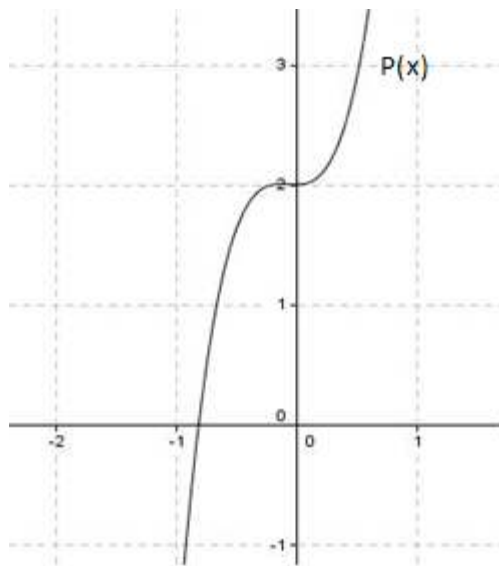
End of Question 13

Question 14 (15 marks) Use a SEPARATE writing booklet

(a)

- (i) The polynomial $P(x) = 5x^3 + x^2 + 2$ has one real root in the interval $-1 < x < 1$. By halving the interval show that the root lies between -1 and -0.75 . 2

- (ii) Given the graph as shown below of the polynomial $P(x) = 5x^3 + x^2 + 2$, briefly justify why Newton's method with an initial approximation of $x = 0$, would be ineffective in improving the approximation of the root. 1



- (b) The polynomial $P(x) = x^3 - 3x^2 + kx + 6$ has roots α, β, γ .

- (i) Find the value of $\alpha + \beta + \gamma$ 1
- (ii) Find the value of $\alpha\beta\gamma$ 1
- (iii) It is known that two of the roots are equal in magnitude but opposite in sign. Find the third root and hence find the value of k . 2

Question 14 continues on page 14

Question 14 (continued)

(c) The two points $P(2ap, ap^2)$ and $Q(2aq, aq^2)$ are on the parabola $x^2 = 4ay$.

- (i) The chord PQ varies in such a way as to always pass through the point $(0, -5a)$ as shown on the diagram below. The equation of the chord PQ is $y - \frac{1}{2}(p + q)x + apq = 0$ (Do NOT prove this).

1

Show that $pq = 5$.

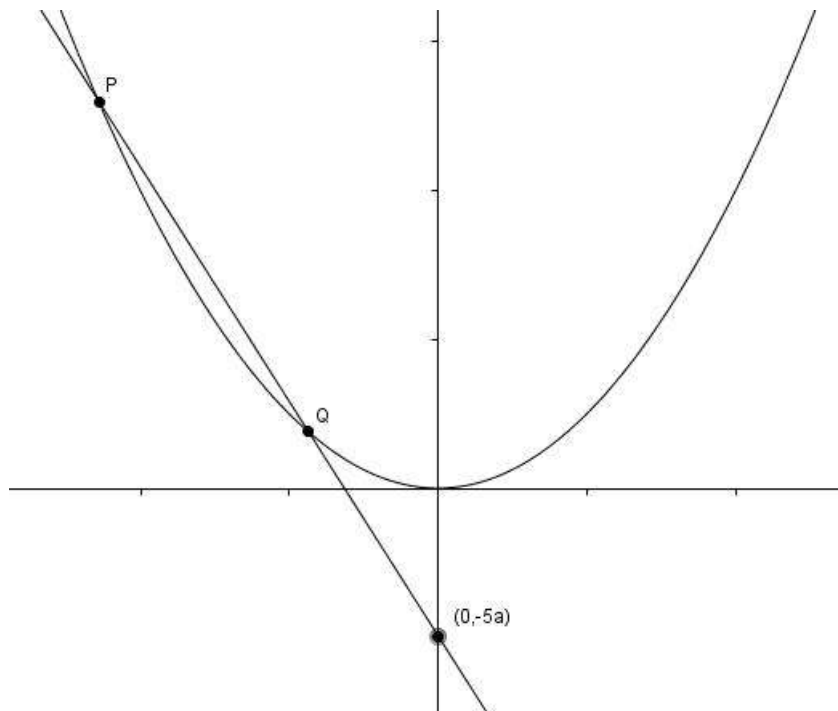
- (ii) The equation of the tangent to $x^2 = 4ay$ at an arbitrary point $(2at, at^2)$ on the parabola is $y = tx - at^2$. (Do NOT prove this).

3

Show that the tangents at the points P and Q meet at R, where R is the point $(a(p + q), apq)$.

- (iii) Hence find the equation of the locus of R.

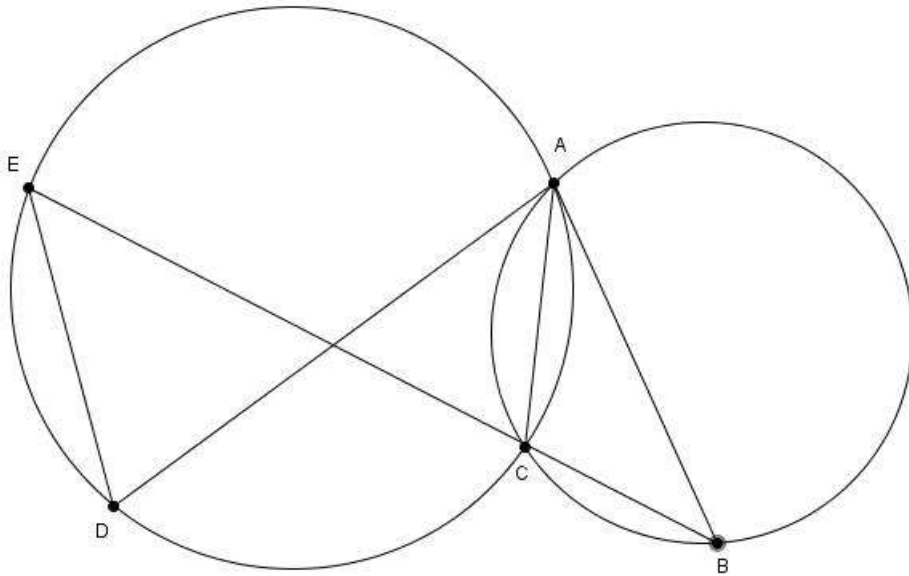
1



Question 14 continues on page 15

Question 14 (continued)

- (d) AD is a tangent to the circle with points A, B and C on its circumference as shown in the diagram below. Copy this diagram. Prove $AB \parallel DE$. 3



End of Question 14

End of Paper ☺

BLANK PAGE



--	--	--	--	--	--	--	--	--

Student Number

Mathematics

Section I – Multiple Choice Answer Sheet

Use this multiple-choice answer sheet for questions 1 – 10. Detach this sheet.

Select the alternative A, B, C or D that best answers the question. Fill in the response oval completely.

Sample: $2 + 4 =$ (A) 2 (B) 6 (C) 8 (D) 9
A ☐ B ☒ C ☐ D ☐

If you think you have made a mistake, put a cross through the incorrect answer and fill in the new answer.

A ☒ B ☒ C ☐ D ☐

If you change your mind and have crossed out what you consider to be the correct answer, then indicate the correct answer by writing the word *correct* and drawing an arrow as follows.

A ☒ B ☒ C ☐ D ☐
correct

- Start Here** →
1. A ☐ B ☐ C ☐ D ☐
 2. A ☐ B ☐ C ☐ D ☐
 3. A ☐ B ☐ C ☐ D ☐
 4. A ☐ B ☐ C ☐ D ☐
 5. A ☐ B ☐ C ☐ D ☐
 6. A ☐ B ☐ C ☐ D ☐
 7. A ☐ B ☐ C ☐ D ☐
 8. A ☐ B ☐ C ☐ D ☐
 9. A ☐ B ☐ C ☐ D ☐
 10. A ☐ B ☐ C ☐ D ☐



Killara
HIGH SCHOOL

Marking Scheme.

Extension 1 - 2016
Half Yearly

A	N	S	W	E	R	S			
---	---	---	---	---	---	---	--	--	--

Student Number

Mathematics

Section I – Multiple Choice Answer Sheet

Use this multiple-choice answer sheet for questions 1 – 10. Detach this sheet.

Select the alternative A, B, C or D that best answers the question. Fill in the response oval completely.

Sample: $2 + 4 =$ (A) 2 (B) 6 (C) 8 (D) 9
A ☐ B ☒ C ☐ D ☐

If you think you have made a mistake, put a cross through the incorrect answer and fill in the new answer.

A ☒ B ☒ C ☐ D ☐

If you change your mind and have crossed out what you consider to be the correct answer, then indicate the correct answer by writing the word *correct* and drawing an arrow as follows.

A ☒ B ☒ C ☐ D ☐
correct

Start
Here



1. A ☒ B ☐ C ☒ D ☐
2. A ☐ B ☐ C ☐ D ☒
3. A ☐ B ☐ C ☒ D ☐
4. A ☐ B ☒ C ☐ D ☐
5. A ☐ B ☐ C ☒ D ☐
6. A ☐ B ☒ C ☐ D ☐
7. A ☒ B ☐ C ☐ D ☐
8. A ☐ B ☐ C ☒ D ☐
9. A ☐ B ☐ C ☐ D ☒
10. A ☒ B ☐ C ☐ D ☐

Multiple Choice

$$1.) \frac{8}{20} = \frac{x}{x+6}$$

$$8(x+6) = 20x$$

$$8x + 48 = 20x$$

$$48 = 12x$$

$$x = 4$$

(A)

$$2.) \cos^6 x - \sin^6 x$$

$$= (\cos^2 x)^3 - (\sin^2 x)^3$$

$$= (\cos^2 x - \sin^2 x)(\cos^4 x + \cos^2 x \sin^2 x + \sin^4 x)$$

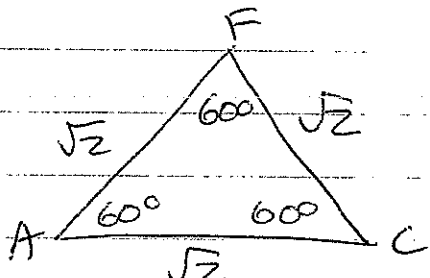
$$= \cos 2x (\cos^4 x + \cos^2 x \sin^2 x + \sin^4 x)$$

(D)

3.) By Pythagoras' Theorem,

$$AC = CF = AF = \sqrt{2}$$

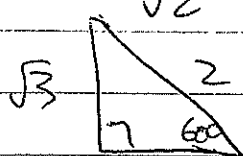
$\triangle ACF$ is equilateral $\therefore \angle AFC = 60^\circ$



$$A = \frac{1}{2} ab \sin C$$

$$= \frac{1}{2} \times \sqrt{2} \times \sqrt{2} \times \sin 60^\circ$$

$$= \frac{1}{2} \times 2 \times \frac{\sqrt{3}}{2}$$



$$\therefore A = \frac{\sqrt{3}}{2} \text{ m}^2$$

(C)

4.)

$$2^{2x} = 5$$

$$\ln 2^{2x} = \ln 5$$

$$2x \ln 2 = \ln 5$$

$$2x = \frac{\ln 5}{\ln 2}$$

$$x = \frac{\ln 5}{2 \ln 2}$$

(B)

or substitute alternatives into original equation.

5.)

$$\int_0^1 6x \sqrt{9-x^2} dx$$

$$= \int_0^1 -3 \sqrt{9-x^2} \cdot -2x dx$$

$$= \int_9^8 -3 \sqrt{u} du$$

$$u = 9 - x^2$$

$$\frac{du}{dx} = -2x$$

$$du = -2x dx$$

$$\text{when } x=0,$$

$$u = 9$$

$$x=1,$$

$$u = 9 - 1$$

$$= 8$$

(C)

Multiple Choice

Q6

$$(x+3)^2 \times \frac{x+1}{x+3} \leq 2 \times (x+3)^2$$

Note $x \neq -3$

$$(x+3)(x+1) \leq 2(x+3)^2$$

$$(x+3)(x+1) + 2(x+3)^2 \leq 0$$

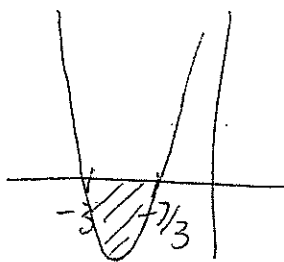
$$(x+3)[(x+1) + 2(x+3)] \leq 0$$

$$(x+3)(x+1+2x+6) \leq 0$$

$$(x+3)(3x+7) \leq 0$$

$\therefore$ x-int's of parabola at $x=-3$ (impossible)

$$\text{and } x = -\frac{7}{3}$$



$$\therefore -3 < x \leq -\frac{7}{3} \quad \textcircled{B}$$

Q7

$$\tan \theta = \left| \frac{m_1 - m_2}{1 + m_1 m_2} \right|$$

$$m_1 = 2, m_2 = -1$$

$$= \left| \frac{2 - (-1)}{1 + 2 \cdot (-1)} \right|$$

$$= \left| \frac{3}{-1} \right| = |-3| = 3$$

$$\therefore \theta = \tan^{-1}(3)$$

$$\therefore \theta = 71^\circ 34' \quad \textcircled{A}$$

Q8

A (-1, 3)

B (4, 12)

$3 \div 2$

x_1, y_1

x_2, y_2

$m \div n$

$$\left(\frac{mx_2 + nx_1}{m+n}, \frac{my_2 + ny_1}{m+n} \right)$$

$$= \left(\frac{3 \times 4 + 2 \times -1}{3+2}, \frac{3 \times 12 + 2 \times -3}{3+5} \right)$$

$$= \left(\frac{12-2}{5}, \frac{36-6}{5} \right) = \left(\frac{10}{5}, \frac{30}{5} \right) = (2, 6) \text{ (C)}$$

Q9

$$\cos 2\theta = \sin \theta$$

$$\cos 2\theta = 1 - 2\sin^2 \theta$$

$$1 - 2\sin^2 \theta = \sin \theta$$

$$2\sin^2 \theta + \sin \theta - 1 = 0$$

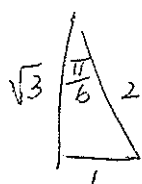
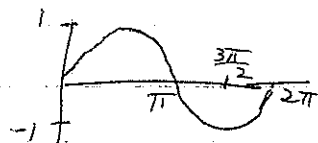
$$\begin{array}{l} 2\sin \theta \\ \sin \theta \end{array} \begin{array}{l} -1 \\ 1 \end{array}$$

$$\therefore (2\sin \theta - 1)(\sin \theta + 1) = 0$$

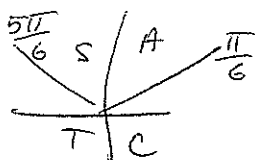
$$\therefore \sin \theta = \frac{1}{2}, \sin \theta = -1$$

$\sin \theta = -1$ is not a solution

in domain $0 \leq \theta \leq \pi$



$$\sin \theta = \frac{1}{2}$$



$\therefore$ solutions are

$$\theta = \frac{\pi}{6}, \frac{5\pi}{6} \text{ (D)}$$

Q10

$$a \sin \theta + b \cos \theta = R \sin(2t + \alpha)$$

$$1 \sin 2t + \sqrt{3} \cos 2t = R \sin(2t + \alpha)$$

$$R = \sqrt{1^2 + (\sqrt{3})^2} = 2$$

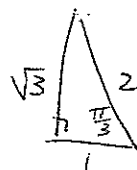
$$\therefore 1 \sin 2t + \sqrt{3} \cos 2t = 2 \sin(2t + \alpha)$$

need α

fast way $\tan \alpha = \frac{b}{a} = \frac{\sqrt{3}}{1}$

$$\alpha = \tan^{-1} \sqrt{3}$$

$$\therefore \alpha = \frac{\pi}{3}$$



$$\therefore 1 \sin 2t + \sqrt{3} \cos 2t = 2 \sin\left(2t + \frac{\pi}{3}\right) \quad \textcircled{A}$$

OR long way

$$1 \sin 2t + \sqrt{3} \cos 2t$$

$$= 2 \left(\frac{1}{2} \sin 2t + \frac{\sqrt{3}}{2} \cos 2t \right)$$

$$= 2 \left(\sin 2t \cdot \frac{1}{2} + \cos 2t \cdot \frac{\sqrt{3}}{2} \right)$$

$$= 2 (\sin 2t \cos \alpha + \cos 2t \sin \alpha)$$

$$= 2 \sin(2t + \alpha)$$

$$\cos \alpha = \frac{1}{2}, \quad \sin \alpha = \frac{\sqrt{3}}{2}$$

$$\tan \alpha = \frac{\sin \alpha}{\cos \alpha} = \frac{\frac{\sqrt{3}}{2}}{\frac{1}{2}} = \sqrt{3}$$

$$\therefore \alpha = \tan^{-1} \sqrt{3} = \frac{\pi}{3}$$

$$= 2 \sin\left(2t + \frac{\pi}{3}\right) \quad \textcircled{A}$$

Question 11

a) $e^{\ln y} = y$

Done poorly. Many students do not realise $e^{\ln(\dots)}$ and $\ln(\dots)$ are opposite operations.

1 mark
Correct answer.

b) $f(x) = \ln(x^2 + e^2)$

$x^2 \geq 0$

$\therefore x^2 + e^2 \geq e^2$

Many could not see this.

1 mark
 $x^2 + e^2 \geq e^2$

$f(x) \geq \ln e^2$
 $\geq 2 \ln e$
 ≥ 2

Simply $y \geq 2$ with no working received 1 mark.
1 mark

$\therefore$ range is $f(x) \geq 2$

Correct range.

c) $\frac{d}{dx} \frac{e^x}{x} = \frac{u}{v}$
 $= \frac{xe^x - e^x}{x^2}$

Well done.

2 marks: Correct derivative.

1 mark:

d) $\frac{d}{dx} \log_5 x^3$
 $= \frac{d}{dx} \frac{\ln x^3}{\ln 5}$
 $= \frac{d}{dx} \frac{3}{\ln 5} \ln x$
 $= \frac{3}{\ln 5} \times \frac{1}{x}$
 $= \frac{3}{x \ln 5}$

Those who realised $\frac{3}{\ln 5}$ is a constant, easily got to the final answer.

Many students attempted quotient rule unsuccessfully.

1 mark
change of base

1 mark
correct derivative

Question 11 (continued)

e.) Limiting sum when $|r| < 1$

(i)

$$r = 2 \sin^2 x$$

$$0 < x < \pi$$

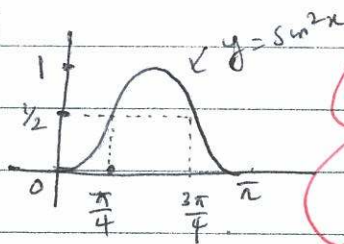
1 mark
for common
ratio between
-1 and 1

$$-1 < 2 \sin^2 x < 1 \quad \checkmark$$

$$-\frac{1}{2} < \sin^2 x < \frac{1}{2}$$

$$0 \leq \sin^2 x < \frac{1}{2}$$

$$\text{Since } \sin^2 x \geq 0$$

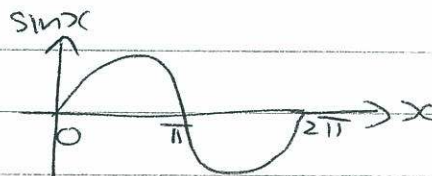
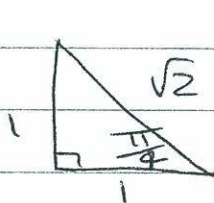


Drawing a
diagram would
have been the
easiest way to
solve this
inequality

$$-\frac{1}{\sqrt{2}} < \sin x < \frac{1}{\sqrt{2}}$$

$$\text{for } 0 < x < \pi, \sin x \geq 0$$

$$\therefore 0 < \sin x < \frac{1}{\sqrt{2}} \quad \checkmark \quad 0 < x < \pi \quad 2 \text{ marks:}$$



Correct solutions
from correct
working.

$$\sin^{-1} 0 < x < \sin^{-1} \frac{1}{\sqrt{2}}$$

$$0 < x < \frac{\pi}{4}$$

Award 1 mark
if one of the
solutions is given
 $0 < x < \pi/4$ or
the other.

$\therefore$ limiting sum exists for

$$0 < x < \frac{\pi}{4}$$

$$\frac{3\pi}{4} < x < \pi$$

Many students solved
the equation $\sin x = \pm \frac{1}{\sqrt{2}}$
and giving the answer
 $-\pi/4 \leq x \leq \pi/4$ or
 $\frac{\pi}{4} \leq x \leq 3\pi/4$. No
marks.

Question 11 (cont.)

e.)

(ii)

$$S_{\infty} = \frac{a}{1-r}$$

$$a = \sin^2 x$$

Well done

$$r = 2\sin^2 x$$

$$S_{\infty} = \frac{\sin^2 x}{1 - 2\sin^2 x}$$

$$= \frac{\sin^2 x}{1 - \sin^2 x - \sin^2 x}$$

$$= \frac{\sin^2 x}{\cos^2 x - \sin^2 x}$$

$$= \frac{\sin^2 x}{\cos 2x}$$

1 mark
expression
for S_{∞}

1 mark
final
answer

Question 11 (cont.)

f.)

(i) 15, 19, 23, 27, ...

An AP with $a = 15$, $d = 4$

$$\begin{aligned} S_n &= \frac{n}{2} (2a + (n-1)d) \\ &= \frac{n}{2} (30 + 4(n-1)) \\ &= \frac{n}{2} (30 + 4n - 4) \\ &= \frac{n}{2} (26 + 4n) \\ &= n (13 + 2n) \\ S_n &= 2n^2 + 13n \end{aligned}$$

Well done

1 mark
correct
expression
in any form.

Some students found
 T_n instead of S_n

$$\begin{aligned} \text{(ii)} \quad \frac{S_n}{n} &= 43 \\ \frac{n(13+2n)}{n} &= 43 \\ 13+2n &= 43 \\ 2n &= 30 \\ n &= 15 \end{aligned}$$

1 mark
 $n = 15$
rows in
the theatre.

$$\begin{aligned} T_n &= a + (n-1)d \\ T_{15} &= 15 + 14 \times 4 \\ &= 71 \text{ seats} \end{aligned}$$

Well done

There are 71 seats in the
last row of the theatre.

1 mark
71 seats
in last row.

Q12a $u = \ln x$

$$\frac{du}{dx} = \frac{1}{x}$$

$$\therefore du = \frac{1}{x} dx$$

$$\therefore \int \frac{1}{x (\ln x)^4} dx = \int \frac{1}{(\ln x)^4} \cdot \frac{1}{x} dx$$

$$= \int \frac{1}{u^4} \cdot du \quad \left. \vphantom{\int \frac{1}{u^4} \cdot du} \right\} \begin{array}{l} (1 \text{ mark}) \\ \text{correct} \\ \text{integral in} \\ \text{terms of } u \end{array}$$

$$= \int u^{-4} du$$

$$= \frac{u^{-3}}{-3} + C \quad \left. \vphantom{\frac{u^{-3}}{-3} + C} \right\} \begin{array}{l} (1 \text{ mark}) \\ \text{correct} \\ \text{integration} \end{array}$$

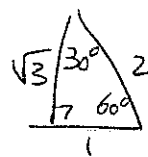
$$= \frac{(\ln x)^{-3}}{-3} + C$$

$$= \frac{-1(\ln x)^{-3}}{3} + C \quad \left. \vphantom{\frac{-1(\ln x)^{-3}}{3} + C} \right\} \begin{array}{l} (1 \text{ final} \\ \text{answer} \\ \text{in either form}) \end{array}$$

$$= \frac{-1}{3(\ln x)^3} + C$$

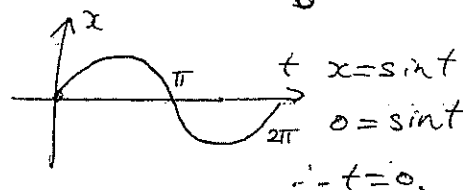
$$\frac{b}{2} \int_0^{\frac{1}{2}} \sqrt{1-x^2} dx$$

$$\text{let } x = \sin t$$



$$\begin{aligned} \text{limits } \frac{1}{2} &= \sin t \\ t &= \sin^{-1} \frac{1}{2} \\ t &= \frac{\pi}{6} \end{aligned}$$

(1 mark
correct
limits)



$$\therefore \int_0^{\frac{1}{2}} \sqrt{1-x^2} dx$$

$$= \int_0^{\frac{\pi}{6}} \sqrt{1-(\sin t)^2} dt \cdot \cos t$$

$$\text{if } x = \sin t$$

$$\frac{dx}{dt} = \cos t$$

$$\therefore dx = \cos t dt$$

$$= \int_0^{\frac{\pi}{6}} \sqrt{\cos^2 t} \cdot \cos t dt$$

$$= \int_0^{\frac{\pi}{6}} \cos t \cdot \cos t dt$$

$$(1 \text{ mark}) = \int_0^{\frac{\pi}{6}} \cos^2 t dt$$

$$\begin{cases} \cos 2t = 2\cos^2 t - 1 \\ 2\cos^2 t = \cos 2t + 1 \\ \cos^2 t = \frac{1}{2}(\cos 2t + 1) \end{cases}$$

$$(1 \text{ mark}) = \int_0^{\frac{\pi}{6}} \frac{1}{2}(\cos 2t + 1) dt$$

$$= \frac{1}{2} \int_0^{\frac{\pi}{6}} \cos 2t + 1 dt$$

$$(1 \text{ mark}) = \frac{1}{2} \left[\frac{1}{2} \sin 2t + t \right]_0^{\frac{\pi}{6}}$$

$$= \frac{1}{2} \left[\left(\frac{1}{2} \sin \frac{\pi}{3} + \frac{\pi}{6} \right) - \left(\frac{1}{2} \sin 0 + 0 \right) \right] = \frac{1}{2} \left(\frac{1}{2} \cdot \frac{\sqrt{3}}{2} + \frac{\pi}{6} \right)$$

$$(1 \text{ mark}) = \frac{\sqrt{3}}{4} + \frac{\pi}{12}$$

C I $\cos(\theta + 2\theta) = \cos\theta \cdot \cos 2\theta - \sin\theta \cdot \sin 2\theta$

Note: $\sin 2\theta = 2\sin\theta \cos\theta$

$$\cos 2\theta = 2\cos^2\theta - 1$$

sub into $\cos\theta \cdot \cos 2\theta - \sin\theta \cdot \sin 2\theta$

$$= \cos\theta(2\cos^2\theta - 1) - \sin\theta \cdot 2\sin\theta \cdot \cos\theta$$

$$= 2\cos^3\theta - \cos\theta - 2\sin^2\theta \cdot \cos\theta \quad (1 \text{ mark})$$

Note: $\sin^2\theta + \cos^2\theta = 1$

$$\sin^2\theta = 1 - \cos^2\theta$$

$$= 2\cos^3\theta - \cos\theta - 2(1 - \cos^2\theta) \cdot \cos\theta$$

$$= 2\cos^3\theta - \cos\theta - 2\cos\theta + 2\cos^3\theta$$

$$= 4\cos^3\theta - 3\cos\theta$$

$\therefore$ as $\cos(\theta + 2\theta) = \cos 3\theta$

$$\cos 3\theta = 4\cos^3\theta - 3\cos\theta \quad (1 \text{ mark})$$

ii $\int 12\cos^3\theta - 9\cos\theta \cdot d\theta$

$$= \int 3(4\cos^3\theta - 3\cos\theta) d\theta$$

$$= 3 \int 4\cos^3\theta - 3\cos\theta d\theta$$

$$= 3 \int \cos 3\theta d\theta \quad \underline{1 \text{ mark}}$$

$$= 3 \cdot \frac{1}{3} \sin 3\theta + C$$

$$= \sin 3\theta + C \quad \underline{1 \text{ mark}}$$

Q12d

$$\left[\begin{array}{l} \int x \sqrt{1-3x} \, dx \\ x \leq \frac{1}{3} \end{array} \right.$$

$$\left\{ \begin{array}{l} \text{let } u = 1-3x \\ \frac{du}{dx} = -3 \\ du = -3dx \\ dx = \frac{du}{-3} \end{array} \right.$$

$$\left[\begin{array}{l} u = 1-3x \\ 3x = 1-u \\ x = \frac{1-u}{3} = \frac{1}{3}(1-u) \end{array} \right.$$

$$\therefore \int x \sqrt{1-3x} \, dx$$

$$= \int \frac{1}{3}(1-u) \cdot \sqrt{u} \cdot \frac{du}{-3}$$

$$= \int -\frac{1}{9}(1-u) \cdot u^{1/2} \, du \quad (1 \text{ mark})$$

$$= -\frac{1}{9} \int u^{1/2} - u^{3/2} \, du$$

$$= -\frac{1}{9} \left(\frac{2u^{3/2}}{3} - \frac{2u^{5/2}}{5} \right) + C \quad (1 \text{ mark})$$

$$= \frac{-2u^{3/2}}{27} + \frac{2u^{5/2}}{45} + C = \frac{-2(1-3x)^{3/2}}{27} + \frac{2(1-3x)^{5/2}}{45} + C$$

(1 mark)

Question 13

$$(i) f(x) = \frac{2x^2}{9-x^2}$$

$$= \frac{2x^2}{(3+x)(3-x)}$$

vertical asymptotes at
 $x = -3, x = 3$

1 mark
 vertical
 asymptotes.

$$f(x) = \frac{2x^2}{9-x^2}$$

$$= \frac{\frac{2x^2}{x^2}}{\frac{9}{x^2} - 1} = \frac{2}{\frac{9}{x^2} - 1}$$

as $x \rightarrow \infty$,

$$f(x) \rightarrow \frac{2}{-1} = -2$$

1 mark
 horizontal
 asymptote.

horizontal asymptote at $y = -2$

$$(ii) f(x) = \frac{2x^2}{9-x^2}$$

$$f(-x) = \frac{2(-x)^2}{9-(-x)^2}$$

$$= \frac{2x^2}{9-x^2}$$

$$= f(x)$$

$\therefore f(x)$ is even.

1 mark

showing
 $f(x)$ is
 even.

Question 13a (cont.)

$$\begin{aligned} \text{(iii)} \quad f(x) &= \frac{2x^2}{9-x^2} = \frac{u}{v} \\ f'(x) &= \frac{v \frac{du}{dx} - u \frac{dv}{dx}}{v^2} \\ &= \frac{(9-x^2)(4x) - 2x^2(-2x)}{(9-x^2)^2} \\ &= \frac{36x - 4x^3 + 4x^3}{(9-x^2)^2} \\ &= \frac{36x}{(9-x^2)^2} \end{aligned}$$

Stationary points at $f'(x) = 0$

$$0 = \frac{36x}{(9-x^2)^2}$$

$$36x = 0$$

$$x = 0$$

$$\text{at } x=0, \quad y = \frac{0}{9-0} = 0$$

$\therefore$ stationary point at $(0,0)$
test its nature.

x	-1	0	1
$f'(x)$	-0.5625	0	0.5625
	\	—	/

$$\text{at } x = -1, \quad f'(x) = \frac{-36}{82} = -0.5625$$

$$\text{at } x = 1, \quad f'(x) = \frac{36}{82} = 0.5625$$

$\therefore$ A minimum at $(0,0)$

There is a minimum stationary point at $(0,0)$.

Some students wrote horizontal asymptotes $x = \pm 3$ which should be $x = \pm 3$

generally well done

1 mark stationary point at $(0,0)$

quite a few did not give values.

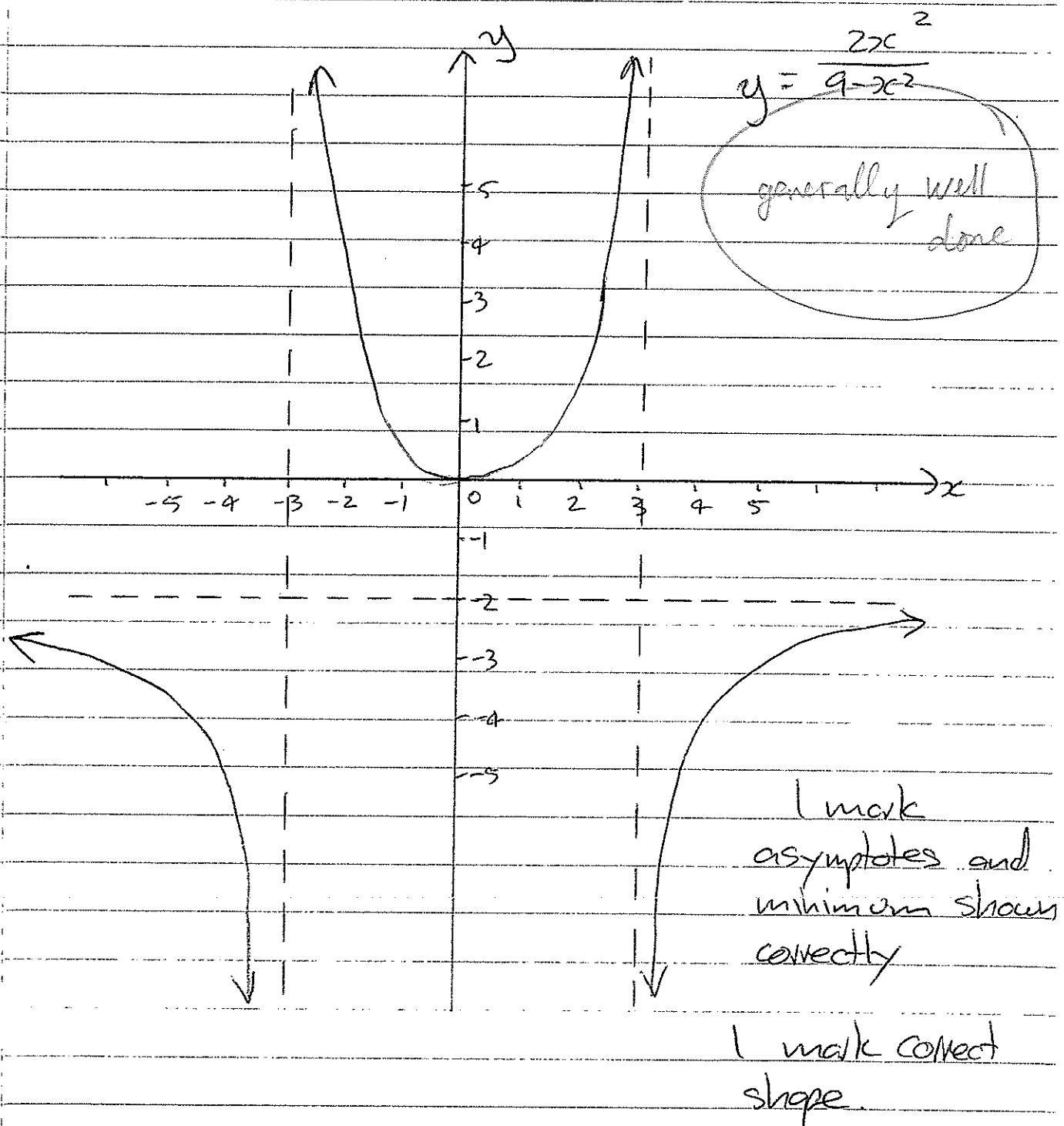
1 mark prove point is a minimum.

Question 13a (cont.)

(iv) even function

asymptotes at $x = -3$, $x = 3$, $y = -2$
minimum at $(0, 0)$

$$\text{at } x = 4, y = \frac{32}{9-16} = \frac{-32}{7} = -4\frac{4}{7}$$
$$x = -4, y = -4\frac{4}{7}$$



Question 13b

(i)

$$\text{In } \triangle ACD, \sin \alpha = \frac{CD}{b}$$

$$CD = b \sin \alpha$$

$$\text{In } \triangle BCD, \sin \beta = \frac{CD}{a}$$

$$CD = a \sin \beta$$

$$\therefore a \sin \beta = b \sin \alpha$$

1 mark
showing
result

$$(ii) \text{ In } \triangle ACD, \cos \alpha = \frac{AD}{b}$$

$$AD = b \cos \alpha$$

$$\text{In } \triangle BCD, \cos \beta = \frac{BD}{a}$$

$$BD = a \cos \beta$$

$$c = AD + BD$$

$$\therefore c = b \cos \alpha + a \cos \beta$$

1 mark
showing
result

Well done

$$(iii) c^2 = 4ab \cos \alpha \cos \beta$$

from part (ii)

$$(b \cos \alpha + a \cos \beta)^2 = 4ab \cos \alpha \cos \beta$$

$$b^2 \cos^2 \alpha + 2ab \cos \alpha \cos \beta + a^2 \cos^2 \beta = 4ab \cos \alpha \cos \beta$$

$$b^2 \cos^2 \alpha - 2ab \cos \alpha \cos \beta + a^2 \cos^2 \beta = 0$$

$$(b \cos \alpha - a \cos \beta)^2 = 0$$

$$b \cos \alpha - a \cos \beta = 0$$

$$b \cos \alpha = a \cos \beta$$

$$b^2 \cos^2 \alpha = a^2 \cos^2 \beta$$

$$b^2 (1 - \sin^2 \alpha) = a^2 (1 - \sin^2 \beta)$$

$$b^2 - b^2 \sin^2 \alpha = a^2 - a^2 \sin^2 \beta$$

$$\text{from (i), } a \sin \beta = b \sin \alpha$$

$$b^2 - b^2 \sin^2 \alpha = a^2 - b^2 \sin^2 \alpha$$

$$b^2 = a^2$$

$$b = a \quad \text{since } a > 0, b > 0$$

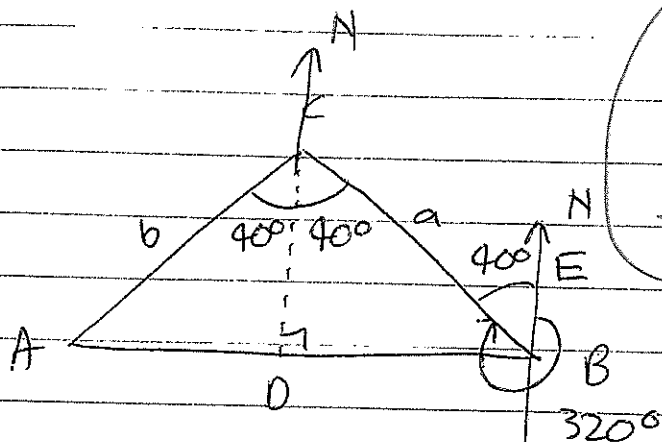
$$\therefore a = b$$

1 mark
final proof

A few found
difficulty in
this part

Question 13b (cont.)

(iv)



quite a few assumed that A is West of B. The answer was correct but the proof was incorrect

$$\angle BCD = \angle EBC = 40^\circ$$

(alternate angles
EB || CD)

from part (iii) $a = b$
 $\therefore \triangle ABC$ is isosceles
 and CD bisects $\angle ACB$
 $\therefore \angle ACD = 40^\circ$
 $\therefore \angle ACB = 80^\circ$

$$v.) AD = \frac{85}{2} = 42.5 \text{ m}$$

In $\triangle ACD$, $\sin 40^\circ = \frac{42.5}{b}$

$$b = \frac{42.5}{\sin 40^\circ}$$

$$= 66.11826... \text{ m}$$

In $\triangle ATC$,

$$\tan 23^\circ = \frac{h}{b}$$

$$h = \tan 23^\circ \times 66.11826...$$

$$= 28.065537...$$

$$= 28.1 \text{ m (1 d.p.)}$$

Generally
Well done

1 mark
for
geometrical
explanation
may be
several
different
ones.

1 mark
calculate
length of
b or a

1 mark
calculate
h

1 mark
correct rounding
to 1 d.p.

14a i

$$P(x) = 5x^3 + x^2 + 2$$

12

$$\begin{aligned} P(-1) &= 5 \times (-1)^3 + (-1)^2 + 2 \\ &= -5 + 1 + 2 \\ &= -2 \end{aligned}$$

$$\begin{aligned} P(1) &= 5 \times 1^3 + 1^2 + 2 \\ &= 8 \end{aligned}$$

Many students did not show that one value was < 0 and one value > 0 and thus the root lies between them

as $P(-1) < 0$ and $P(1) > 0$

root lies between -1 and 1

$$P\left(\frac{-1+1}{2}\right) = P(0) = 5 \times 0^3 + 0^2 + 2 = 2$$

as $P(-1) < 0$ and $P(0) > 0$

root lies between -1 and 0

need the conclusion

$$P\left(\frac{-1+0}{2}\right) = P\left(-\frac{1}{2}\right) = 5 \times \left(-\frac{1}{2}\right)^3 + \left(-\frac{1}{2}\right)^2 + 2$$

$$= 5 \times -\frac{1}{8} + \frac{1}{4} + 2$$

$$= -\frac{5}{8} + \frac{2}{8} + 2$$

$$= -\frac{3}{8} + \frac{16}{8} = \frac{13}{8}$$

(1 mark)
calculate
 $P(-1)$, $P(1)$
 $P(0)$, $P\left(-\frac{1}{2}\right)$
and concludes

$\therefore$ as $P\left(-\frac{1}{2}\right) > 0$ and $P(-1) < 0$

root lies between -1 and $-\frac{1}{2}$

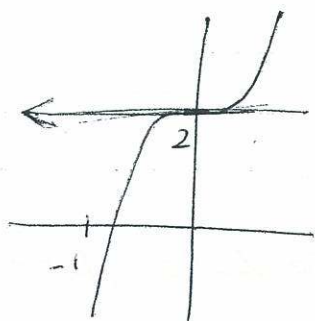
$$P\left(\frac{-1 + -\frac{1}{2}}{2}\right) = P\left(-\frac{3}{4}\right) = 5 \times \left(-\frac{3}{4}\right)^3 + \left(-\frac{3}{4}\right)^2 + 2$$

$$= 5 \times -\frac{27}{64} + \frac{9}{16} + 2 = \frac{29}{64}$$

$\therefore$ as $P(-1) < 0$ and $P(0.75) > 0$
the root lies between
-1 and 0.75

(1 mark)
calculates
 $P(0.75)$
and $P(-1)$

ii



very poorly done
→ need to be specific

The tangent at $x=0$
is a turning point (as shown)
This will not meet
the x-axis, so it will be
impossible for Newton's
method to work.

(other valid answers related
to the tangent should gain a mark)

b i

$$P(x) = x^3 - 3x^2 + Kx + 6$$

$$\alpha + \beta + \gamma = \frac{-b}{a} = \frac{-(-3)}{1}$$

$$\therefore \alpha + \beta + \gamma = 3$$

ii

$$\alpha\beta\gamma = \frac{-d}{a} = \frac{-6}{1} = -6$$

iii

let $\beta = -\alpha$ (equal in magnitude, opposite in sign)

$$\alpha + \beta + \gamma = 3$$

$$\alpha + -\alpha + \gamma = 3$$

$$\therefore \gamma = 3$$

(1 mark)

$$2\beta\gamma = -6$$

$$\gamma = 3$$

$$\therefore 2\beta \cdot 3 = -6$$

$$2\beta = -2$$

$$\text{as } \beta = -\alpha$$

$$\alpha \cdot -\alpha = -2$$

$$-\alpha^2 = -2$$

$$\alpha^2 = 2$$

$$\therefore \alpha = \pm\sqrt{2}$$

$$\text{so } \alpha = \sqrt{2}, \beta = -\sqrt{2}$$

$$2\beta + 2\gamma + \beta\gamma = \frac{c}{a} = \frac{k}{1} = k$$

$$\text{as } \beta = -\alpha$$

$$\alpha \cdot -\alpha + 2\gamma - \alpha\gamma = k$$

$$-\alpha^2 = k$$

$$-(\sqrt{2})^2 = k$$

$$\therefore k = -2$$

(1 mark)

c i

$$y - \frac{1}{2}(p+q)x + apq = 0$$

$$\text{sub } x=0 \quad y + apq = 0$$

$$y = -apq$$

we know y-intercept = $-5a$

$$\therefore \frac{-5a}{-a} = \frac{-apq}{-a}$$

$$\therefore pq = 5$$

many had a
very long winded
approach $\rightarrow$ it was
only worth 1 mark!

ii at P

tangent is $y = px - ap^2$ ①

3

at Q

tangent is $y = qx - aq^2$ ②

(1 mark)

solve simultaneously

$$\begin{aligned} \text{①} - \text{②} \quad 0 &= px - qx - ap^2 + aq^2 \\ 0 &= x(p - q) - a(p^2 - q^2) \end{aligned}$$

$$\frac{x(p - q)}{(p - q)} = \frac{a(p - q)(p + q)}{(p - q)}$$

$$\therefore x = a(p + q)$$

(1 mark)

sub into $y = px - ap^2$

$$y = p \cdot a(p + q) - ap^2$$

$$y = ap^2 + apq - ap^2$$

$$\therefore y = apq$$

(1 mark)

$\therefore$ point of intersect R is $(a(p + q), apq)$

iii

$$y = apq \quad (\text{from part ii})$$

$$pq = 5 \quad (\text{from part i})$$

$$\begin{aligned} \therefore y &= a \times pq \\ &= a \times 5 \end{aligned}$$

$$\therefore y = 5a \quad (\text{horizontal line through } 5a)$$

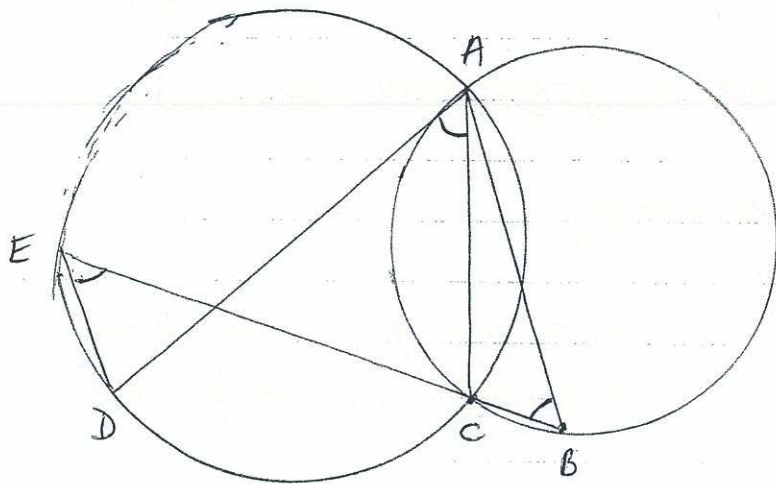
(1 mark for $y = 5a$ with working)

many students got this but didn't recognise it as a complete solution

Note: $y = 5a$ doesn't count in itself. You have to state $y = 5a$ with working.

d

Many students used abbreviation
→ this is very risky and
not advised
→ reasons need to be much
better expressed



(1 mark)
 $\angle DAC = \angle ABC$ (angle between a tangent and
a chord through the point of contact
is equal to the angle in the alternate
segment)

(1 mark)
 $\angle DEC = \angle DAC$ (angles in the same segment
of a circle are equal)

$\therefore \angle DEC = \angle DAC = \angle ABC$ (from above)

$\therefore \angle DEC = \angle ABC$

(1 mark)
note: $\angle EBA = \angle ABC$ (same angle), similarly $\angle DEC = \angle DEB$
 $\therefore$ As alternate angles ^{equal}, $\angle DEB = \angle EBA$
this means the lines DE and AB
cut by the transversal EB are $\parallel$.
 $\therefore AB \parallel DE$